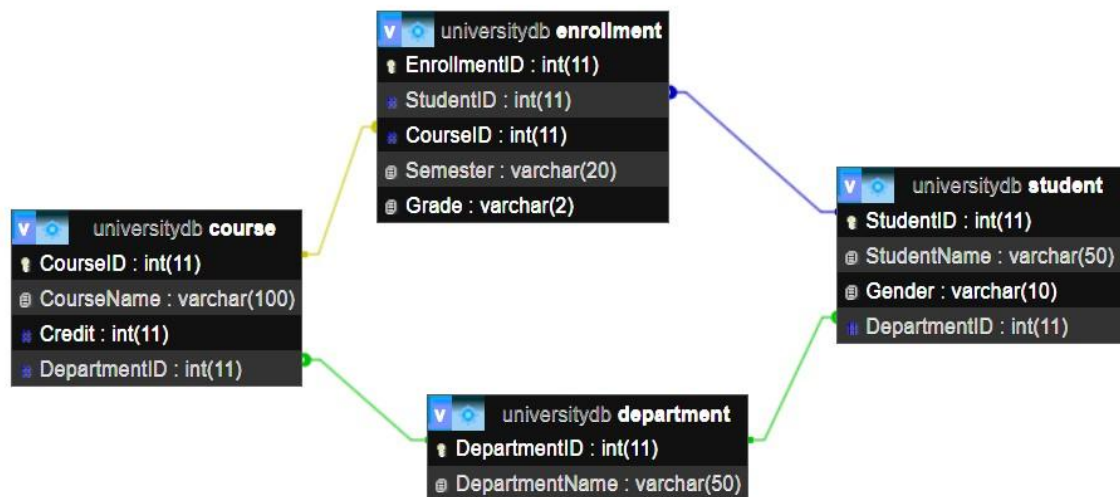


Title: INNER JOIN Operations in MySQL

Objective: The objective of this task is to use INNER JOIN to , retrieve data from multiple tables. By completing this task I'll learn how to design a relational database and write SQL queries to analyze data effectively.

ERD :



Basic INNER JOIN

Q1. Display the student name and department name.

This query joins the Student and Department tables to display each student's name with the department name.

```
MariaDB [UniversityDB]> SELECT StudentName, DepartmentName  
-> FROM Student  
-> INNER JOIN Department  
-> ON Student.DepartmentID=Department.DepartmentID;
```

StudentName	DepartmentName
Ayan	CSE
Sarah	CSE
Nafis	EEE
Mim	EEE
Rafi	BBA
Nusrat	BBA
Hasan	English
Tania	English

```
8 rows in set (0.000 sec)
```

Q2. Display the course name and department name.

This displays every course together with the department that offers it.

```
MariaDB [UniversityDB]> SELECT CourseName, DepartmentName  
-> FROM Course  
-> INNER JOIN Department  
-> ON Course.DepartmentID=Department.DepartmentID;
```

CourseName	DepartmentName
Database Systems	CSE
Data Structures	CSE
Programming Fundamentals	CSE
Circuit Analysis	EEE
Digital Electronics	EEE
Marketing Principles	BBA
Business Communication	BBA
English Literature	English

```
8 rows in set (0.000 sec)
```

```
MariaDB [UniversityDB]> |
```

Q3. Display the student name, course name, and semester.

This query combines three tables

```
MariaDB [UniversityDB]> SELECT StudentName, CourseName, Semester  
-> FROM Enrollment  
-> INNER JOIN Student  
-> ON Enrollment.StudentID=Student.StudentID  
-> INNER JOIN Course  
-> ON Enrollment.CourseID=Course.CourseID;
```

StudentName	CourseName	Semester
Ayan	Database Systems	Spring 2026
Ayan	Data Structures	Spring 2026
Sarah	Database Systems	Spring 2026
Sarah	Programming Fundamentals	Fall 2025
Nafis	Circuit Analysis	Spring 2026
Nafis	Digital Electronics	Fall 2025
Mim	Digital Electronics	Spring 2026
Rafi	Marketing Principles	Spring 2026
Nusrat	Business Communication	Fall 2025
Hasan	English Literature	Spring 2026
Tania	English Literature	Fall 2025
Tania	Business Communication	Spring 2026

12 rows in set (0.000 sec)

```
MariaDB [UniversityDB]> D
```

Q4. Display the student name, course name, and grade.

This query shows the grades obtained by students in each course.

```
MariaDB [UniversityDB]> SELECT StudentName, CourseName, Grade  
-> FROM Enrollment  
-> INNER JOIN Student  
-> ON Enrollment.StudentID=Student.StudentID  
-> INNER JOIN Course  
-> ON Enrollment.CourseID=Course.CourseID;
```

StudentName	CourseName	Grade
Ayan	Database Systems	A
Ayan	Data Structures	A-
Sarah	Database Systems	B+
Sarah	Programming Fundamentals	A
Nafis	Circuit Analysis	A
Nafis	Digital Electronics	B
Mim	Digital Electronics	A-
Rafi	Marketing Principles	B+
Nusrat	Business Communication	A
Hasan	English Literature	B
Tania	English Literature	A
Tania	Business Communication	A

12 rows in set (0.001 sec)

Q5. Display all enrollment details with student names instead of student IDs.

This query replaces the Student ID with the student's name

```
MariaDB [UniversityDB]> SELECT EnrollmentID, StudentName, CourseID, Semester, Grade
-> FROM Enrollment
-> INNER JOIN Student
-> ON Enrollment.StudentID=Student.StudentID;
```

EnrollmentID	StudentName	CourseID	Semester	Grade
1	Ayan	201	Spring 2026	A
2	Ayan	202	Spring 2026	A-
3	Sarah	201	Spring 2026	B+
4	Sarah	203	Fall 2025	A
5	Nafis	204	Spring 2026	A
6	Nafis	205	Fall 2025	B
7	Mim	205	Spring 2026	A-
8	Rafi	206	Spring 2026	B+
9	Nusrat	207	Fall 2025	A
10	Hasan	208	Spring 2026	B
11	Tania	208	Fall 2025	A
12	Tania	207	Spring 2026	A

12 rows in set (0.000 sec)

INNER JOIN with WHERE

Q6. Display students belonging to the 'CSE' department.

This query finds the names of students who are in the CSE department.

```
MariaDB [UniversityDB]> SELECT StudentName
-> FROM Student
-> INNER JOIN Department
-> ON Student.DepartmentID=Department.DepartmentID
-> WHERE DepartmentName='CSE';
```

StudentName
Ayan
Sarah

2 rows in set (0.000 sec)

Q7. Display all students enrolled in the Database Systems course.

```
MariaDB [UniversityDB]> SELECT StudentName
-> FROM Enrollment
-> INNER JOIN Student
-> ON Enrollment.StudentID=Student.StudentID
-> INNER JOIN Course
-> ON Enrollment.CourseID=Course.CourseID
-> WHERE CourseName='Database Systems';
```

StudentName
Ayan
Sarah

```
2 rows in set (0.001 sec)
```

This finds all students who are taking the Database Systems course.

Q8. Display students who received grade A.

```
MariaDB [UniversityDB]> SELECT StudentName
-> FROM Enrollment
-> INNER JOIN Student
-> ON Enrollment.StudentID=Student.StudentID
-> WHERE Grade='A';
```

StudentName
Ayan
Sarah
Nafis
Nusrat
Tania
Tania

```
6 rows in set (0.000 sec)
```

This query shows all students who got an A grade.

Q9. Display students enrolled in the 'Spring 2026' semester.

```
MariaDB [UniversityDB]> SELECT StudentName
-> FROM Enrollment
-> INNER JOIN Student
-> ON Enrollment.StudentID=Student.StudentID
-> WHERE Semester='Spring 2026';
```

StudentName
Ayan
Ayan
Sarah
Nafis
Mim
Rafi
Hasan
Tania

8 rows in set (0.000 sec)

Displays students who enrolled during the Spring 2026 semester.

Q10. Display female students with their department names.

```
MariaDB [UniversityDB]> SELECT StudentName, DepartmentName
-> FROM Student
-> INNER JOIN Department
-> ON Student.DepartmentID=Department.DepartmentID
-> WHERE Gender='Female';
```

StudentName	DepartmentName
Sarah	CSE
Mim	EEE
Nusrat	BBA
Tania	English

4 rows in set (0.000 sec)

Lists all female students together with their department names.

INNER JOIN with ORDER BY

Q11. Display all students and their departments sorted by student name.

```
MariaDB [UniversityDB]> SELECT StudentName, DepartmentName
-> FROM Student
-> INNER JOIN Department
-> ON Student.DepartmentID=Department.DepartmentID
-> order BY StudentName;
```

StudentName	DepartmentName
Ayan	CSE
Hasan	English
Mim	EEE
Nafis	EEE
Nusrat	BBA
Rafi	BBA
Sarah	CSE
Tania	English

8 rows in set (0.000 sec)

This query sorts the students alphabetically by their names.

Q12. Display

all courses and departments sorted by course name.

```
MariaDB [UniversityDB]> SELECT CourseName, DepartmentName
-> FROM Course
-> INNER JOIN Department
-> ON Course.DepartmentID=Department.DepartmentID
-> ORDER BY CourseName;
```

CourseName	DepartmentName
Business Communication	BBA
Circuit Analysis	EEE
Data Structures	CSE
Database Systems	CSE
Digital Electronics	EEE
English Literature	English
Marketing Principles	BBA
Programming Fundamentals	CSE

8 rows in set (0.000 sec)

This query displays all courses in alphabetical order along with departments.

students and their grades sorted in descending

Q13. Display
order of grade.

```
MariaDB [UniversityDB]> SELECT StudentName, Grade  
-> FROM Enrollment  
-> INNER JOIN Student  
-> ON Enrollment.StudentID=Student.StudentID  
-> ORDER BY Grade DESC;
```

StudentName	Grade
Rafi	B+
Sarah	B+
Nafis	B
Hasan	B
Mim	A-
Ayan	A-
Ayan	A
Tania	A
Nafis	A
Tania	A
Sarah	A
Nusrat	A

12 rows in set (0.001 sec)

Displays student grades and sorts them in descending order.

Q14. Display

student-course information sorted by semester.

```
MariaDB [UniversityDB]> SELECT StudentName, CourseName, Semester
-> FROM Enrollment
-> INNER JOIN Student
-> ON Enrollment.StudentID = Student.StudentID
-> INNER JOIN Course
-> ON Enrollment.CourseID = Course.CourseID
-> ORDER BY Semester;
```

StudentName	CourseName	Semester
Sarah	Programming Fundamentals	Fall 2025
Nafis	Digital Electronics	Fall 2025
Nusrat	Business Communication	Fall 2025
Tania	English Literature	Fall 2025
Ayan	Database Systems	Spring 2026
Mim	Digital Electronics	Spring 2026
Hasan	English Literature	Spring 2026
Sarah	Database Systems	Spring 2026
Tania	Business Communication	Spring 2026
Ayan	Data Structures	Spring 2026
Nafis	Circuit Analysis	Spring 2026
Rafi	Marketing Principles	Spring 2026

12 rows in set (0.000 sec)

Q15. Display

Student course information sorted alphabetically by semester.

students sorted by department name and then student name.

```
MariaDB [UniversityDB]> SELECT StudentName, DepartmentName
-> FROM Student
-> INNER JOIN Department
-> ON Student.DepartmentID = Department.DepartmentID
-> ORDER BY DepartmentName, StudentName;
```

StudentName	DepartmentName
Nusrat	BBA
Rafi	BBA
Ayan	CSE
Sarah	CSE
Mim	EEE
Nafis	EEE
Hasan	English
Tania	English

3 rows in set (0.001 sec)

Sorts students first by department name and then alphabetically by student name

INNER JOIN with Aggregate Functions

Q16. Count the number of students in each department.

```
MariaDB [UniversityDB]> SELECT DepartmentName,  
-> COUNT(Student.StudentID) AS TotalStudents  
-> FROM Department  
-> INNER JOIN Student  
-> ON Department.DepartmentID = Student.DepartmentID  
-> GROUP BY DepartmentName;  
+-----+-----+  
| DepartmentName | TotalStudents |  
+-----+-----+  
| BBA            | 2             |  
| CSE            | 2             |  
| EEE            | 2             |  
| English        | 2             |  
+-----+-----+  
4 rows in set (0.000 sec)
```

This query counts how many students are in each department.

Q17. Count the number of students enrolled in each course.

```
MariaDB [UniversityDB]> SELECT CourseName,  
-> COUNT(Enrollment.StudentID) AS TotalStudents  
-> FROM Course  
-> INNER JOIN Enrollment  
-> ON Course.CourseID = Enrollment.CourseID  
-> GROUP BY CourseName;
```

CourseName	TotalStudents
Business Communication	2
Circuit Analysis	1
Data Structures	1
Database Systems	2
Digital Electronics	2
English Literature	2
Marketing Principles	1
Programming Fundamentals	1

8 rows in set (0.000 sec)

This query counts the number of students enrolled in each courses.

Q18. Find the average credit of courses offered by each department.

```
MariaDB [UniversityDB]> SELECT DepartmentName,  
-> AVG(Credit) AS AverageCredit  
-> FROM Department  
-> INNER JOIN Course  
-> ON Department.DepartmentID = Course.DepartmentID  
-> GROUP BY DepartmentName;
```

DepartmentName	AverageCredit
BBA	3.0000
CSE	3.3333
EEE	3.5000
English	3.0000

4 rows in set (0.000 sec)

This query calculates the average course credit offered by each department

Q19. Find the maximum course credit in each department.

```
MariaDB [UniversityDB]> SELECT DepartmentName,  
-> MAX(Credit) AS MaximumCredit  
-> FROM Department  
-> INNER JOIN Course  
-> ON Department.DepartmentID = Course.DepartmentID  
-> GROUP BY DepartmentName;
```

DepartmentName	MaximumCredit
BBA	3
CSE	4
EEE	4
English	3

4 rows in set (0.000 sec)

This query displays the maximum course credit offered by each department,

Q20. Count the number of enrollments in each semester.

```
MariaDB [UniversityDB]> SELECT Semester,  
-> COUNT(EnrollmentID) AS TotalEnrollments  
-> FROM Enrollment  
-> GROUP BY Semester;
```

Semester	TotalEnrollments
Fall 2025	4
Spring 2026	8

2 rows in set (0.000 sec)

This query counts the total number of enrollments in each semester.

INNER JOIN with GROUP BY & HAVING

Q21. Display departments having more than 2 students.

```
h] MariaDB [UniversityDB]> SELECT DepartmentName,  
h]     -> COUNT(Student.StudentID) AS TotalStudents  
che]     -> FROM Department  
che]     -> INNER JOIN Student  
:ql]     -> ON Department.DepartmentID = Student.DepartmentID  
:ql]     -> GROUP BY DepartmentName  
     -> HAVING COUNT(Student.StudentID) > 2;  
Empty set (0.000 sec)
```

My tables doesn't have departments having more than 2 students.

Q22. Display courses having more than 3 enrolled students.

```
Empty set (0.000 sec)  
MariaDB [UniversityDB]> SELECT CourseName,  
     -> COUNT(Enrollment.StudentID) AS TotalStudents  
     -> FROM Course  
     -> INNER JOIN Enrollment  
     -> ON Course.CourseID = Enrollment.CourseID  
     -> GROUP BY CourseName  
     -> HAVING COUNT(Enrollment.StudentID) > 3;  
Empty set (0.000 sec)
```

Empty for my case.

Q23. Display semesters with more than 5 enrollments.

```
MariaDB [UniversityDB]> SELECT Semester,  
-> COUNT(EnrollmentID) AS TotalEnrollments  
-> FROM Enrollment  
-> GROUP BY Semester  
-> HAVING COUNT(EnrollmentID) > 5;
```

Semester	TotalEnrollments
Spring 2026	8

1 row in set (0.000 sec)

This query shows semesters that have more than five enrollments.

Q24. Display departments offering more than 2 courses.

```
MariaDB [UniversityDB]> SELECT DepartmentName,  
-> COUNT(Course.CourseID) AS TotalCourses  
-> FROM Department  
-> INNER JOIN Course  
-> ON Department.DepartmentID = Course.DepartmentID  
-> GROUP BY DepartmentName  
-> HAVING COUNT(Course.CourseID) > 2;
```

DepartmentName	TotalCourses
CSE	3

1 row in set (0.000 sec)

Displays departments that offer more than two courses.

Q25. Display grades obtained by more than one student.

```
MariaDB [UniversityDB]> SELECT Grade,  
-> COUNT(*) AS TotalStudents  
-> FROM Enrollment  
-> GROUP BY Grade  
-> HAVING COUNT(*) > 1;
```

Grade	TotalStudents
A	6
A-	2
B	2
B+	2

4 rows in set (0.000 sec)

This query displays grades that have been obtained by more than one student.

Conclusion:

This lab helped me learn the use of **INNER JOIN** to retrieve information from multiple tables , while also using **WHERE**, **ORDER BY**, **GROUP BY**, and **HAVING** clauses with **INNER JOIN**.